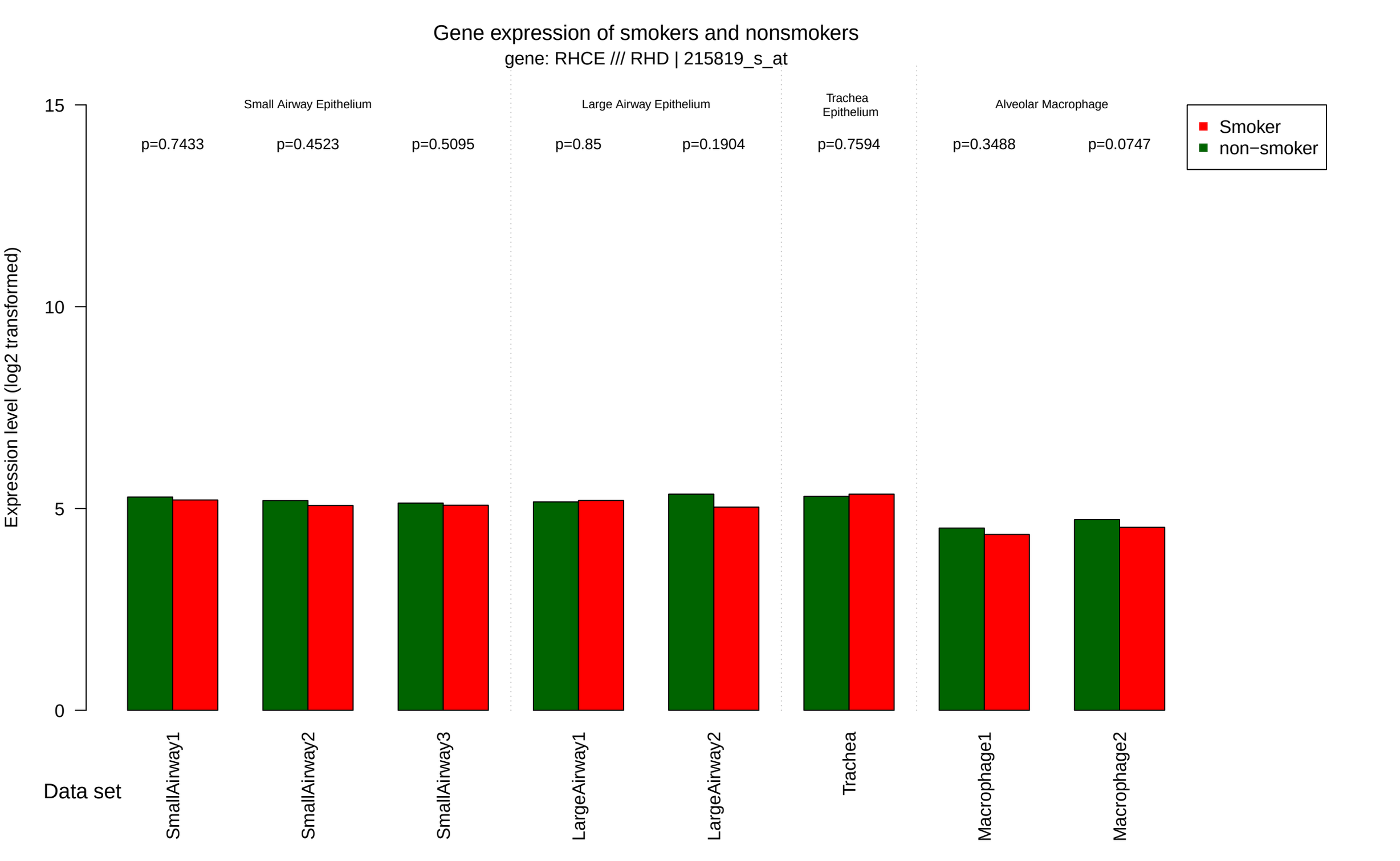
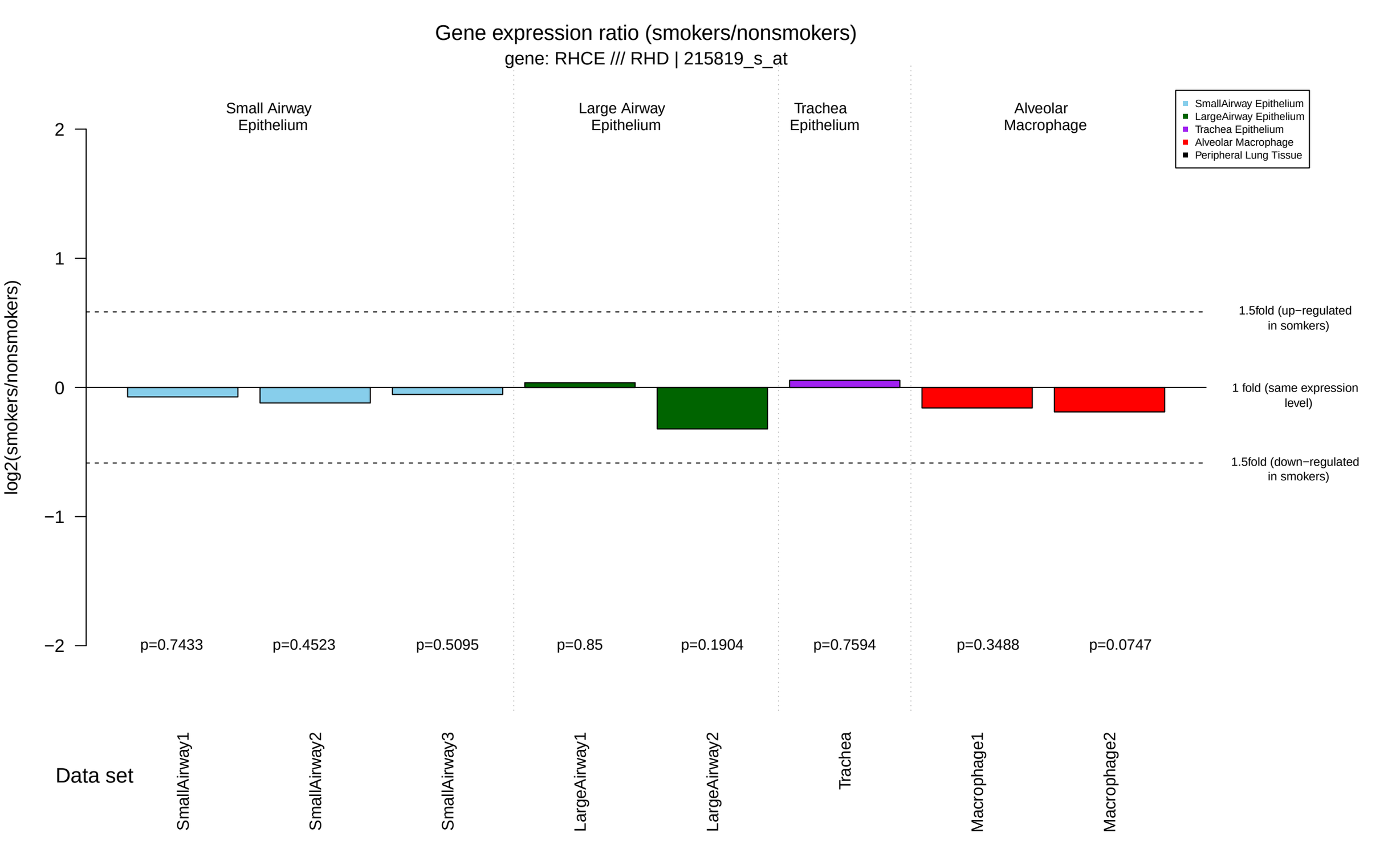
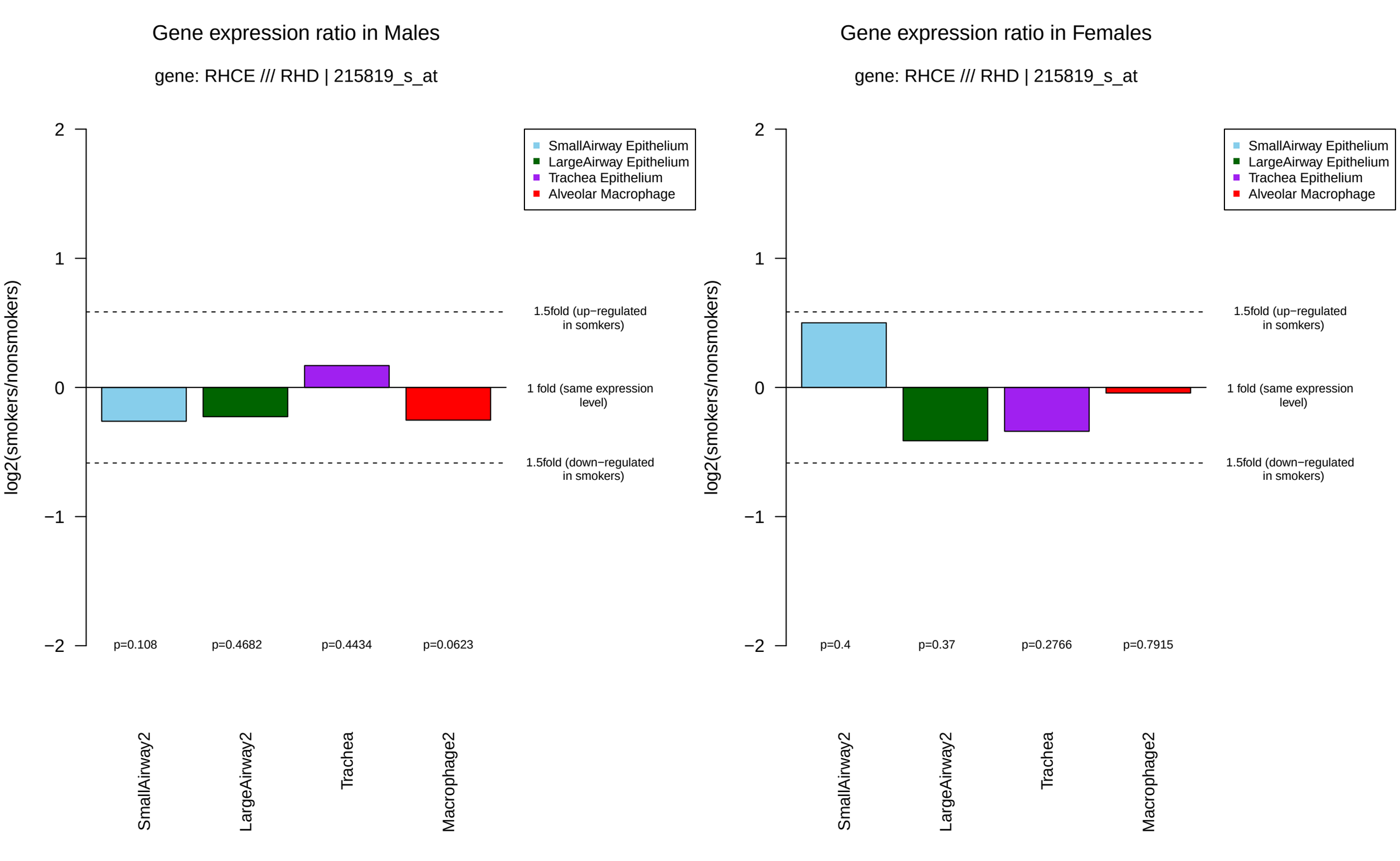
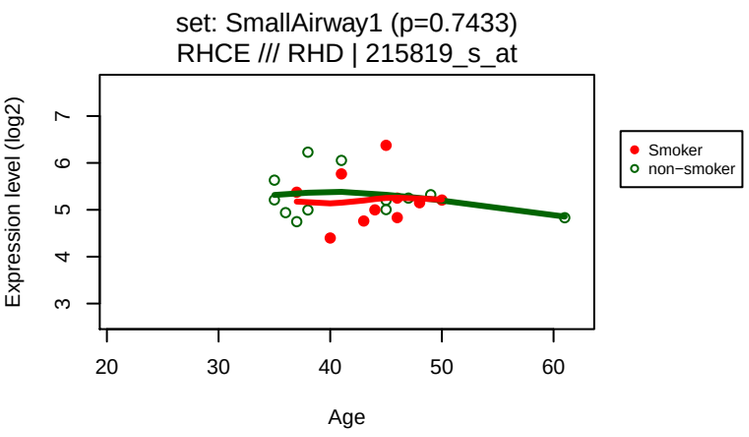
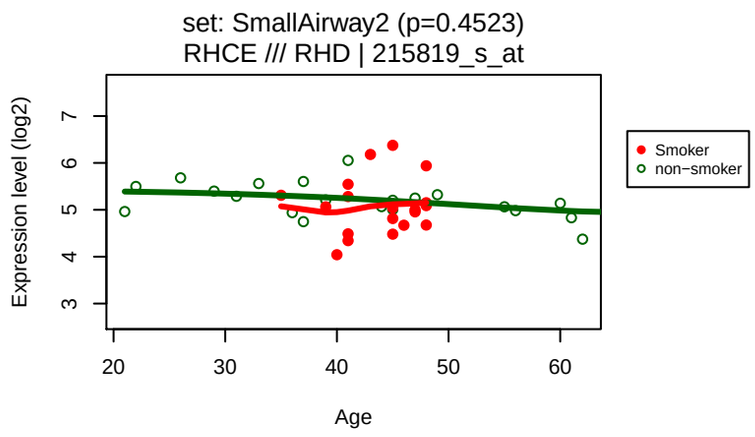
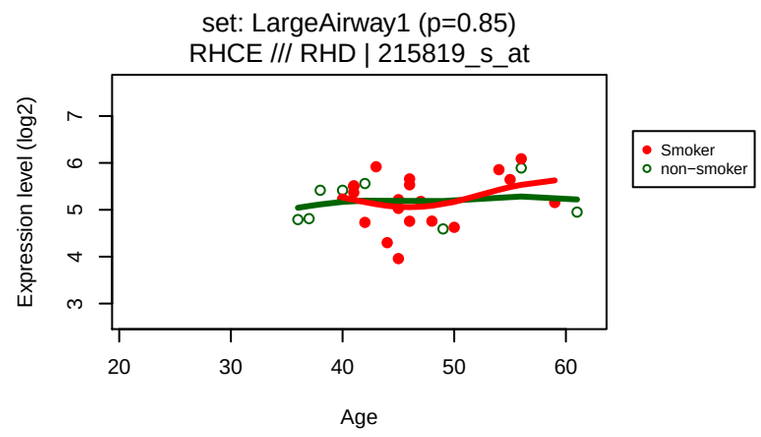

Gene expression of smokers and nonsmokers
gene: RHCE /// RHD | 215819_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7433
p=0.4523
p=0.5095
p=0.85
p=0.1904
p=0.7594
p=0.3488
p=0.0747
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

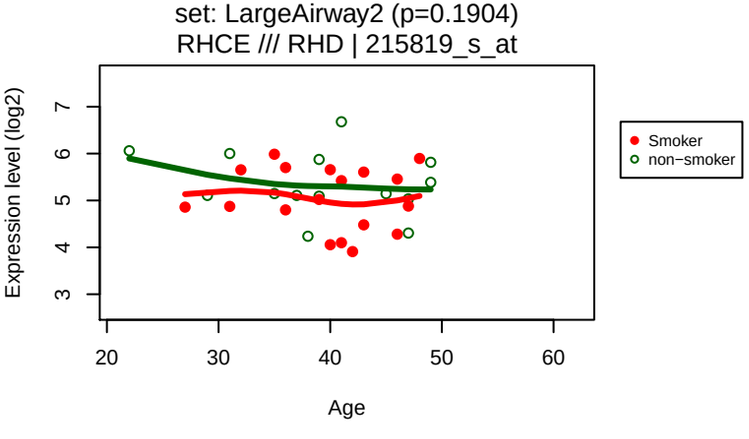

Gene expression ratio (smokers/nonsmokers)
gene: RHCE /// RHD | 215819_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7433
p=0.4523
p=0.5095
p=0.85
p=0.1904
p=0.7594
p=0.3488
p=0.0747
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

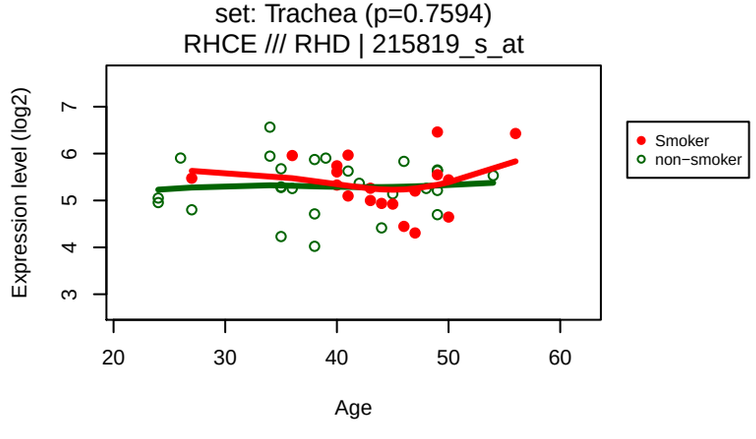

Gene expression ratio in Males
gene: RHCE /// RHD | 215819_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.108
p=0.4682
p=0.4434
p=0.0623
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHCE /// RHD | 215819_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4
p=0.37
p=0.2766
p=0.7915
SmallAirway2
LargeAirway2
Trachea
Macrophage2

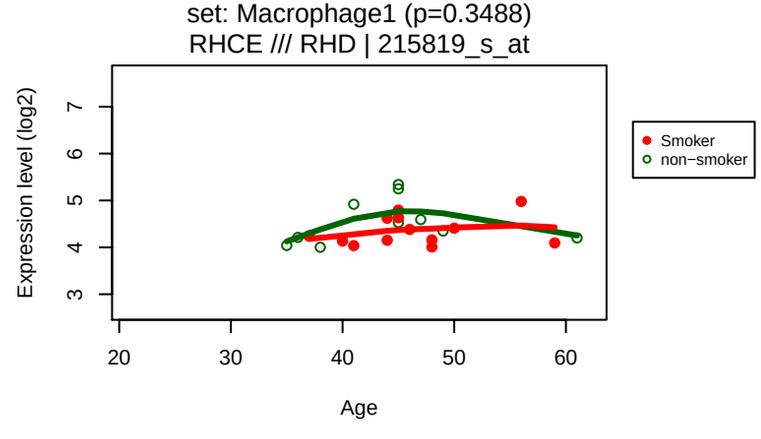

set: SmallAirway1 (p=0.7433)
RHCE /// RHD | 215819_s_at
Expression level (log2)
Age
Smoker
non−smoker

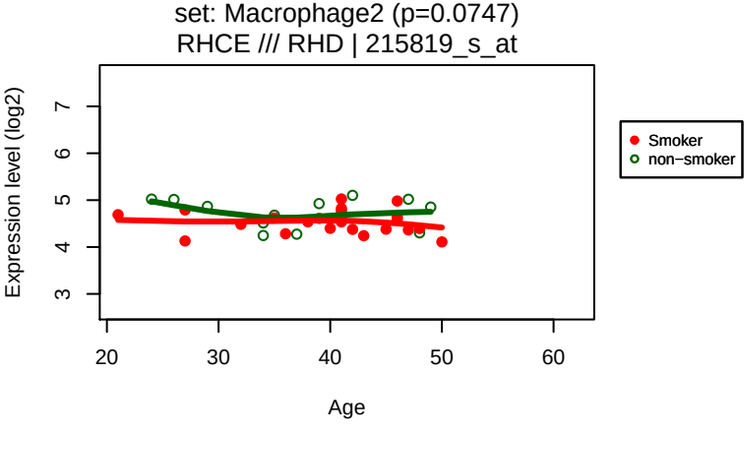

set: SmallAirway2 (p=0.4523)
RHCE /// RHD | 215819_s_at
Expression level (log2)
Age
Smoker
non−smoker

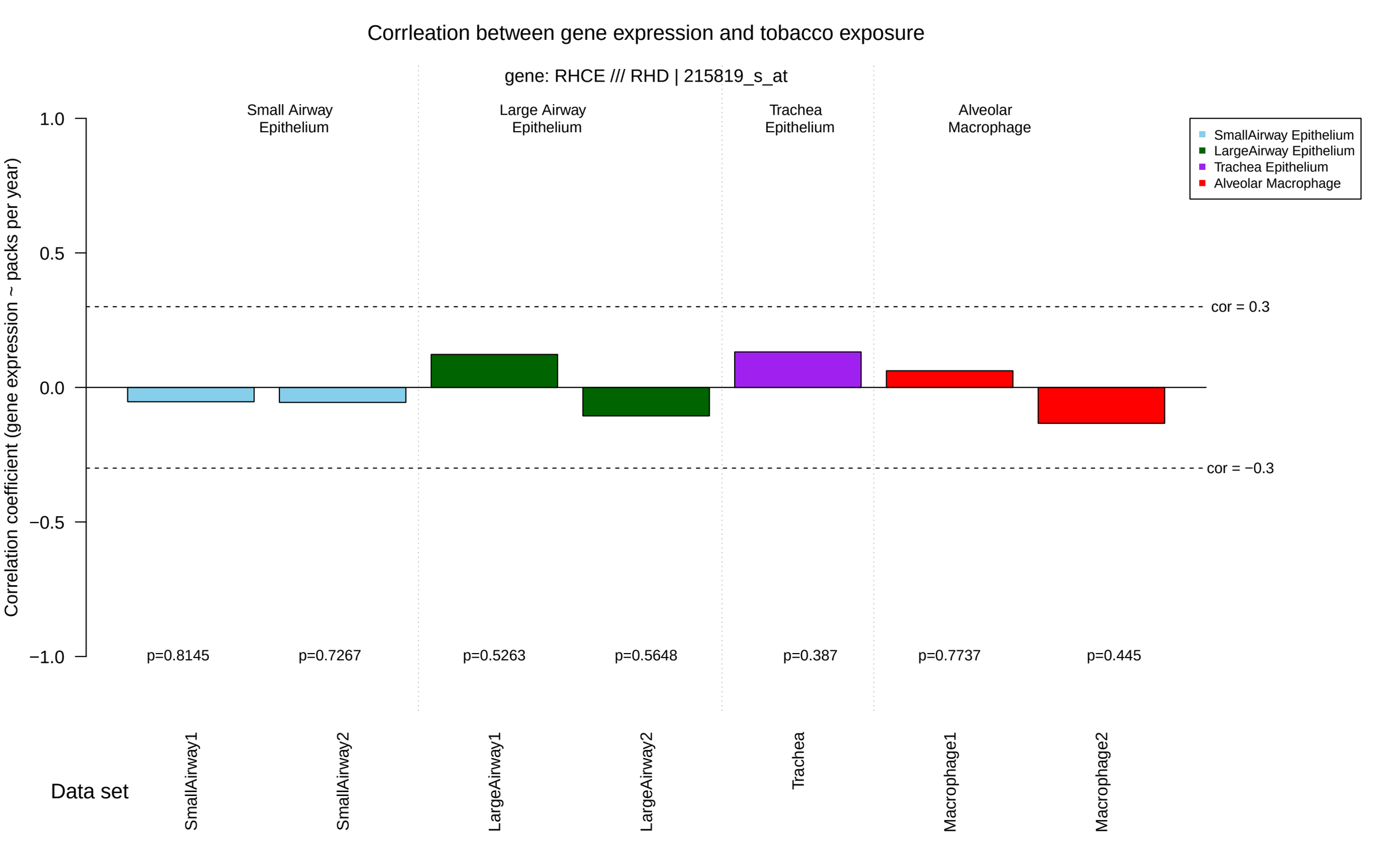

set: LargeAirway1 (p=0.85)
RHCE /// RHD | 215819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1904)
RHCE /// RHD | 215819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7594)
RHCE /// RHD | 215819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3488)
RHCE /// RHD | 215819_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0747)
RHCE /// RHD | 215819_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHCE /// RHD | 215819_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8145
p=0.7267
p=0.5263
p=0.5648
p=0.387
p=0.7737
p=0.445
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2